



UNIVERSITY OF AMSTERDAM

PRODUCT VISION AND RESEARCH

VirtuoPet

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1 Value proposition

VirtuoPet helps people with ASD build healthy habits and understand how their daily routines affect their wellbeing, motivated by intrinsic empathy fostered by caring for a virtual pet.

2 The user & the problem

People with ASD often struggle establishing a clear daily routine, which is where a mobile application can be of much assistance. Many health and wellness apps which support routine management already exist, however they are often not designed with ASD in mind.

Some research has, for example, explored gamified approaches for this user base (Kim et al. 2024). However, some findings suggest that this does not motivate people with ASD and may even cause anxiety (Asselt and Roke 2025).

The biggest issue for people on the spectrum is not a lack of knowledge or willpower. Autistic people often know what they should do, yet struggle to initiate and follow through; there is a disconnect between intention and action (Kim et al. 2024). An app that simply reminds users of their habits does not suffice. It needs to actively support the step from intention to behavior.

Abstract and delayed rewards do not motivate. Knowing that you will be better off in a hypothetical future, or seeing a simple number tick up (Kim et al. 2024), does not motivate a large portion of autistic people enough to stick to habits (Asselt and Roke 2025). This is the core problem virtuoPet aims to solve.

3 Design rationale

VirtuoPet motivates the user through evoking empathy for a digital pet. We turn logging habits into a fun and easy process, through which we aim to make initiation and follow-through easier. Unlike points or streaks, caring for a virtual creature provides immediate and emotionally meaningful feedback, something that resonates with a large portion of autistic users who are not motivated by abstract or delayed rewards.

This core mechanic is supported by a set of evidence-based design principles, outlined in the following section, that tailor the app specifically to the needs of autistic users.

4 UX principles

The following principles guide the UX design of VirtuoPet.

4.1 Avoid gamification

Research has shown that gamification amplifies stress for autistic users rather than providing motivation, partly because extrinsic reward systems disrupt daily routines (Asselt and Roke 2025). VirtuoPet, therefore, avoids streaks, leaderboards, and point systems entirely. The animated virtual pet is the only exception to this. The rest of the app is kept intuitive and minimalist.

4.2 Encouragement over punishment

There is a very fine line between encouragement and discouragement when giving feedback (Mazefsky et al. 2020). That's why, for the well-being of the user, we need to strike this balance in a well thought out way.

Consequently, we decided that the pet can only exist in neutral and happy states, never in a negative one. The app, for example, does not display the pet as being in a famished state when the user has not logged their meal in a while.

All communication within the app supports a low-stress environment in the form of quotes and feedback, according to the NICE Guidelines (National Institute for Health and Care Excellence 2021). These guidelines emphasize that a concrete and structured approach is needed to avoid ambiguity. Our quotes are straightforward and predictable, and not commanding. All quotes are respectful and non-judgmental. According to the guideline's warning against excessive use of metaphors, the quotes do not contain any euphemisms or figurative language.

4.3 Predictability and consistency

For people on the spectrum, a predictable and consistent interface is essential to prevent confusion and frustration. In practice, this means that colors should always carry the same meaning throughout the app. The specific color matters less than its consistent application, as individual preferences for color schemes vary considerably (Asselt and Roke 2025).

Accordingly, familiar user interface patterns should be used when available. Modern apps often reuse patterns, such as navigation bars, that users have come to (subconsciously) expect in apps. This is even more important for people with autism because discrepancies between expectation and reality are experienced more negatively by them (Cruys et al. 2014).

In VirtuoPet, this translates to a consistent font, a consistent color for primary actions and the reuse of familiar layout components. For example, the toolbar at the bottom follows a pattern from numerous popular apps. Furthermore, icons correspond with expected actions: the plus icon adds to the habit log, and the user icon opens the profile overview.



Figure 1: VirtuoPet toolbar with familiar layout

4.4 Minimize cognitive load

Hick's law is a widely known UX design principle, which states that the time it takes to make a decision increases with the number and complexity of choices (Yablonski 2020). Based on this, we should aim to minimize the number of options a user has available on any screen. This is especially relevant for autistic users, who are particularly susceptible to cognitive overload (Cendrine 2019). Too many choices can trigger overwhelm and avoidance, directly undermining the initiation and follow-through that the app aims to support.

Therefore, VirtuoPet minimizes the number of options available to ones that are directly useful. On the home screen, there are only 3 interactive options: logging a habit, viewing your statistics, and viewing your user settings. When the user clicks on one of these, it opens a popup with more detail, allowing for easy access to more functionality without overwhelming the user on the initial screen.

4.5 Customizability

Autistic users have highly varied needs and preferences, meaning a one-size-fits-all approach is unlikely to remain effective across the diversity of the user base (Asselt and Roke 2025). VirtuoPet therefore gives users full control over which habit modules are active, which pet they identify with most, and what their personal goals are, rather than imposing global defaults. This can be seen in figure 2.

4.6 Measurability

As a direct extension of the predictability principle, everything shown to the user must be transparent and traceable. Vague composite scores or a mood bar derived from user input are explic-

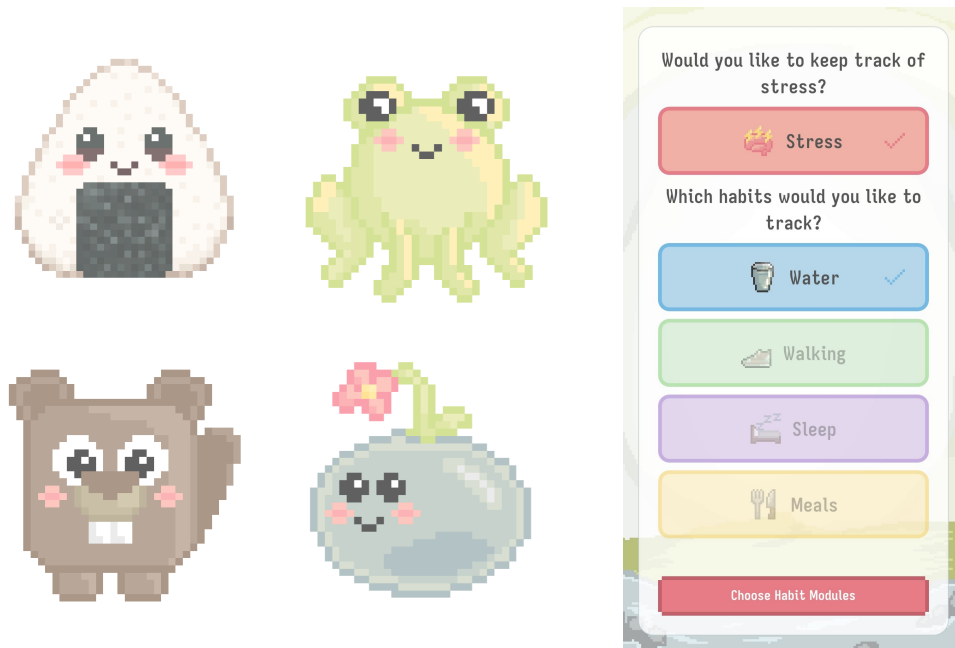


Figure 2: Selectable pets and modules

itly avoided, as they leave too much room for interpretation and undermine the user’s trust in what the app is telling them (Asselt and Roke 2025). Every metric displayed must be directly traceable to concrete, understandable data.

5 Stress module

To help users understand how their daily routines affect their wellbeing, VirtuoPet includes a stress module based on the EMA version of the Perceived Stress Scale (Murray et al. 2023). Users are asked three questions about their day, rated on a five-point slider. These items measure whether the user felt nervous or stressed, whether they felt unable to cope with their tasks, and whether difficulties felt insurmountable. The total score ranges from 3 to 15 and is divided into three levels: low (3–7), moderate (8–11), and high (12–15).

The user is able to view a historic timeline of how stressed they felt. This also displays which of your goals you reached that day. This combination allows the user to correlate their habits and wellbeing over time, making the relationship between their habits and their wellbeing observable and measurable.

It is worth noting several limitations of this approach. First, the scoring division into three levels is a design choice inspired by PSS-10 conventions (Cohen, Kamarck, and Mermelstein 1983), not a validated threshold specific to the EMA version. Second, the original PSS-EMA was designed to measure stress over the past 30 minutes. VirtuoPet adapts this to a daily timeframe, which may affect the psychometric properties of the instrument. Third, the questionnaire was validated in a sample of young adults in Switzerland, which limits generalizability. The stress score should therefore be understood as an indication of experienced stress, not as a clinical measurement.

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